

AP CALCULUS AB - UNIT 1

1.15 Connecting Limits at Infinity and Horizontal Asymptotes

Lesson Overview

- Evaluate limits as $x \rightarrow \infty$ and $x \rightarrow -\infty$.
- Determine horizontal asymptotes from end limits.
- Use degree comparison, dominant powers, radicals, conjugates, and growth hierarchy.
- Analyze the two ends independently and handle $\sqrt{x^2} = |x|$ correctly.

Key Knowledge and Vocabulary

Right end. $x \rightarrow \infty$ asks about large positive inputs.

Left end. $x \rightarrow -\infty$ asks about large negative inputs.

Horizontal asymptote. If an end limit is L , then $y = L$ is a horizontal asymptote for that end.

Rational degree rule. Lower numerator degree gives 0; equal degrees give leading-coefficient ratio; higher numerator degree is usually unbounded.

Radical warning. $\sqrt{x^2} = |x|$, equal to x on the right and $-x$ on the left.

Explanation and Strategy

Step 1. For rational functions divide by the highest power in the denominator or compare leading terms.

Step 2. For radical differences, rationalize before comparing dominant terms.

Step 3. A curve may cross a horizontal asymptote; an asymptote describes end behavior, not a barrier.

Solved Examples

Worked Example: Equal degrees

Evaluate $\lim_{x \rightarrow \pm\infty} \frac{3x^2 - x}{2x^2 + 5}$.

Solution. Dividing by x^2 gives a limit of $3/2$ at both ends, so $y = 3/2$ is a horizontal asymptote.

Worked Example: Different radical ends

Evaluate $\lim_{x \rightarrow \pm\infty} \frac{\sqrt{x^2 + 1}}{x}$.

Solution. Write $\sqrt{x^2 + 1} = |x|\sqrt{1 + 1/x^2}$. The right limit is 1 and the left limit is -1 .

Worked Example: Conjugate at infinity

Evaluate $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 4x} - x)$.

Solution. Rationalize: $\frac{4x}{\sqrt{x^2 + 4x} + x} = \frac{4}{\sqrt{1 + 4/x} + 1} \rightarrow 2$.

Leveled Practice

Easy Level

Foundational fluency. Focus on notation, definitions, and direct procedures.

1. Evaluate $\lim_{x \rightarrow \infty} \frac{3x+1}{x-4}$.

2. Evaluate $\lim_{x \rightarrow -\infty} \frac{2x^2-5}{x^2+7}$.

3. Evaluate $\lim_{x \rightarrow \infty} \frac{x+1}{x^2+1}$.

4. State the horizontal asymptote of $\frac{5x-2}{2x+3}$.

Medium Level

Apply the idea in one or two connected steps.

5. Evaluate $\lim_{x \rightarrow \pm\infty} \frac{5x^3-x}{2x^3+1}$.

6. Evaluate $\lim_{x \rightarrow \pm\infty} \frac{4-3x^2}{x^3+1}$.

7. Evaluate both end limits of $x^2/(2x-1)$.

8. Evaluate $\lim_{x \rightarrow \infty} \frac{e^x}{1+e^x}$ and $\lim_{x \rightarrow -\infty} \frac{e^x}{1+e^x}$.

Hard Level

Connect representations, conditions, and justification.

9. Evaluate $\lim_{x \rightarrow \pm\infty} \frac{2x^2+1}{\sqrt{4x^4+3}}$.

10. Evaluate both end limits of $x/\sqrt{x^2+9}$.

11. Evaluate $\lim_{x \rightarrow \infty} (\sqrt{x^2+5} - x)$.

12. Evaluate $\lim_{x \rightarrow -\infty} (\sqrt{x^2+5} - x)$.

Difficult Level

Use nonroutine structure and precise mathematical reasoning.

13. Evaluate $\lim_{x \rightarrow \infty} x(\sqrt{x^2+1} - x)$.

14. Evaluate $\lim_{x \rightarrow \infty} (3x^3 - \sqrt{9x^6 + x^2})$.

15. Find a, b so $f(x) = \frac{ax+b}{2x-1}$ has horizontal asymptote $y = 3$ and passes through $(0, -4)$.

Challenge Level

Synthesize, construct, generalize, or prove.

16. Construct a function with right horizontal asymptote $y = 2$ and left horizontal asymptote $y = -3$.

17. Prove that a rational function can have at most one finite horizontal asymptote, the same at both ends, when numerator and denominator have equal degree.

AP-Style Practice

Multiple-Choice Practice – No Calculator

Choose the best answer. Use exact values unless a decimal approximation is requested.

1. $\lim_{x \rightarrow \infty} \frac{4x^2+1}{2x^2-3x}$ equals

(A) 0

(C) 4

(B) 2

(D) $+\infty$

2. $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2+1}}{x}$ equals

(A) -1

(C) 1

(B) 0

(D) $+\infty$

3. If $\lim_{x \rightarrow \infty} f(x) = 5$, then

(A) $x = 5$ is a vertical asymptote

(C) $f(x)$ never equals 5

(B) $y = 5$ is a right-end horizontal asymptote

(D) $f(5) = 0$

Multiple-Choice Practice – Calculator Required

Choose the best answer. Use exact values unless a decimal approximation is requested.

4. A calculator table for $f(x) = \frac{2x^3+1}{x^3-4}$ at large positive and negative inputs approaches

(A) -2

(C) 2

(B) 0

(D) $+\infty$

5. A graphing calculator suggests different horizontal levels at the two ends of $x/\sqrt{x^2+4}$. They are

(A) 0 and 0

(C) -1 on the left and 1 on the right

(B) 1 and 1

(D) -2 and 2

Free-Response Practice – No Calculator

Let $f(x) = \frac{3x^2 - 2x + 1}{\sqrt{9x^4 + 4}}$.

(a) Find $\lim_{x \rightarrow \infty} f(x)$.

[3 points]

(b) Find $\lim_{x \rightarrow -\infty} f(x)$.

[2 points]

(c) State the horizontal asymptote.

[1 points]

Free-Response Practice – Calculator Required

A saturation model is $P(t) = \frac{1200}{1 + 8e^{-0.3t}}$ for $t \geq 0$.

(a) Use a calculator to estimate $P(20)$ and $P(40)$. [2 points]

(b) Find $\lim_{t \rightarrow \infty} P(t)$ and interpret it. [2 points]

(c) Explain why reaching the horizontal asymptote at a finite time is not required. [2 points]